

6 hours on examples/handout

Problem 1 (ZCOE2550)

Using a similar argument as in 2.1,

$$\sum_{k=1}^n \mu(k) \left\lfloor \frac{n}{k} \right\rfloor = \sum_{m=1}^n \sum_{k|m} \mu(k)$$

I don't have much practice writing this type of problem. Can you check that all notation after this is acceptable?

We evaluate $\sum_{k|m} \mu(k)$. Let the prime factorization of $m = p_1^{e_1} \dots p_i^{e_i}$ for $e_1, \dots, e_i > 0$. We can define $S = \{p_1, \dots, p_i\}$. We define the function $\pi(s)$ gives the product of elements of a set s with $\pi(\emptyset) = 1$. Since μ returns zero when passed any non-square-free number,

$$\sum_{k|m} \mu(k) = \sum_{\text{subsets } s \text{ of } S} \mu(\pi(s))$$

We let $f(s)$ return 1 if the number of elements in a set s is even, and -1 otherwise. (I'm worried all these functions might be confusing)

$$= \sum_{\text{subsets } s \text{ of } S} f(s)$$

We let N be the number of elements in S ,

$$= \sum_{i=0}^N (-1)^i \binom{N}{i}$$

By binomial theorem,

$$= \begin{cases} 1 & \text{if } N = 0 \\ 0 & \text{if } N > 0 \end{cases}$$

Since $N = 0 \Leftrightarrow m = 1$, (not specifying the reason or the fact that m is positive integer)

$$= \begin{cases} 1 & \text{if } m = 1 \\ 0 & \text{if } m > 1 \end{cases}$$

Therefore,

$$\sum_{k=1}^n \mu(k) \left\lfloor \frac{n}{k} \right\rfloor = \sum_{m=1}^n \sum_{k|m} \mu(k) = 1$$

Problem 2 (SNAKE2)

We look at generating function,

$$\begin{aligned}
F(X) &= \sum_{n \geq 0} \sum_{k \geq 0} \binom{2k}{k} \binom{n}{k} \left(-\frac{1}{4}\right)^k x^n \\
&= \sum_{k \geq 0} \binom{2k}{k} \left(-\frac{1}{4}\right)^k \sum_{n \geq k} \binom{n}{k} x^n
\end{aligned}$$

We let $m := n - k$.

$$\begin{aligned}
&= \sum_{k \geq 0} \binom{2k}{k} \left(-\frac{1}{4}\right)^k x^k \sum_{m \geq 0} \binom{m+k}{k} x^m \\
&= \sum_{k \geq 0} \binom{2k}{k} \left(-\frac{1}{4}\right)^k x^k (1-x)^{-k-1} \\
&= \frac{1}{1-x} \cdot \sum_{k \geq 0} \binom{2k}{k} \left(-\frac{x}{4-4x}\right)^k \\
&= \frac{1}{1-x} \cdot \frac{1}{\sqrt{1+\frac{x}{1-x}}} = \frac{1}{1-x} \cdot \frac{1}{\sqrt{\frac{1}{1-x}}} \\
&= \frac{1}{\sqrt{1-x}} \\
&= \sum_{k \geq 0} \binom{2k}{k} \left(\frac{x}{4}\right)^k
\end{aligned}$$

Therefore, the answer is $\frac{\binom{2n}{n}}{4^n}$.

Problem 4 (Stanford 2012)

We see,

$$\begin{aligned}
&\sum_{n \geq 1} \frac{1}{n^3(n+1)^3} \\
&= \sum_{n \geq 1} \frac{6n^2 - 3n + 1}{n^3} - \frac{6n^2 + 15n + 10}{(n+1)^3} \\
&= \sum_{n \geq 1} \frac{6n^2 - 3n + 1}{n^3} - \frac{6(n+1)^2 + 3(n+1) + 1}{(n+1)^3} \\
&= -3 \left(\sum_{n \geq 1} \frac{1}{n^2} + \frac{1}{(n+1)^2} \right) + 6 \left(\sum_{n \geq 1} \frac{1}{n} - \frac{1}{n+1} \right) + \sum_{n \geq 1} \frac{1}{n^3} - \frac{1}{(n+1)^3}
\end{aligned}$$

$$\begin{aligned}
&= -6 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) + 3 + 6 + 1 \\
&= 10 - \pi^2
\end{aligned}$$

Problem 7 (10AMO5)

Note that $\frac{2}{i \cdot (i+1) \cdot (i+2)} = \frac{1}{i} - \frac{2}{i+1} + \frac{1}{i+2}$. Therefore,

$$\begin{aligned}
2S_q &= \frac{2}{2 \cdot 3 \cdot 4} + \frac{2}{5 \cdot 6 \cdot 7} + \dots + \frac{2}{q(q+1)(q+2)} \\
&= \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) + \left(\frac{1}{5} - \frac{2}{6} + \frac{1}{7} \right) + \dots + \left(\frac{1}{q} - \frac{2}{q+1} + \frac{1}{q+2} \right) \\
&= \left(\frac{1}{2} + \dots + \frac{1}{q+2} \right) - \left(\frac{3}{3} + \frac{3}{6} + \dots + \frac{3}{q+1} \right) \\
&= \left(\frac{1}{2} + \dots + \frac{1}{\frac{3p-1}{2}} \right) - \left(\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{\frac{p-1}{2}} \right) \\
&= \left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{\frac{3p-1}{2}} \right) - 1
\end{aligned}$$

For some integers m, n ,

$$\Leftrightarrow \frac{m}{n} = \frac{1}{p} - 2S_q = 1 - \left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} + \frac{1}{p+1} + \dots + \frac{1}{\frac{3p-1}{2}} \right)$$

We focus on the cases where $\gcd(n, p) = 1$. Note that if we prove $m - n \mid p$ in that case we'd get $\gcd(m, p) = 1$ as well. From there, we can multiply both top and bottom by the inverse of $\gcd(m, n)$ mod p to get that $m - n \mid p$ when $\frac{m}{n}$ is reduced as well to be done. (Could this have been worded better? Was it necessary to say of this?)

We require that $\gcd(n, p) = 1$.

$$\Leftrightarrow 1 - \frac{m}{n} = \frac{n-m}{n} = \left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} \right) + \left(\frac{1}{p+1} + \dots + \frac{1}{\frac{3p-1}{2}} \right)$$

To prove $n - m \mid p$, we can prove the numerator of $\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} + \frac{1}{p+1} + \dots + \frac{1}{\frac{3p-1}{2}}$ once it's simplified is divisible by p .

We can let $n = \left(\frac{p+1}{2} \cdot \frac{p+3}{2} \cdot \dots \cdot (p-1) \right) \cdot \left((p+1) \cdot \dots \cdot \frac{3p-1}{2} \right)$. As p is prime, clearly it is relatively prime to each of the factors so $\gcd(p, n) = 1$. We have,

$$m = \left(\frac{p+1}{2} \cdot \frac{p+3}{2} \cdot \dots \cdot (p-1) \right) \cdot \left((p+1) \cdot \dots \cdot \frac{3p-1}{2} \right) \cdot \left(\left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} \right) + \left(\frac{1}{p+1} + \dots + \frac{1}{\frac{3p-1}{2}} \right) \right)$$

Since m is an integer, we can use modular inverses instead,

$$\begin{aligned} m &\equiv \left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} \right) + \left(\frac{1}{p+1} + \dots + \frac{1}{\frac{3p-1}{2}} \right) \pmod{p} \\ &\equiv \left(\frac{1}{\frac{p+1}{2}} + \frac{1}{\frac{p+3}{2}} + \dots + \frac{1}{p-1} \right) + \left(\frac{1}{1} + \dots + \frac{1}{\frac{p-1}{2}} \right) \pmod{p} \\ &\equiv 1^{-1} + 2^{-1} + \dots + (p-1)^{-1} \pmod{p} \end{aligned}$$

Using the sum of powers modulo p lemma, since -1 is not a multiple of $p-1$, we are done.

$$\equiv 0 \pmod{p}$$

11 clubs; 7.5 hours

Problem 9 (1802PFA)

$$\begin{aligned} &= \int_0^1 \int_0^{\sqrt{x}} \frac{2xy}{1-y^4} dy dx \\ &= \int_0^1 \int_{y^2}^1 \frac{2xy}{1-y^4} dx dy \\ &= \int_0^1 \frac{y}{1-y^4} \left(\int_{y^2}^1 2x dx \right) dy \\ &= \int_0^1 y \cdot \frac{1-y^4}{1-y^4} dy \\ &= \frac{1}{2} \end{aligned}$$

Problem 11 (95IMO6)

We look at the generating function,

$$F(x, y) = \prod_{j=1}^{2p} (1 + x^j y)$$

Note that the coefficient of $x^m y^n$ is the number of subsets of $\{1, \dots, 2p\}$ with size n and sum of elements m . We start by calculating N the number of subsets with size and sum divisible by p .

Using the roots of unity filter,

$$N = \frac{1}{p^2} \cdot \sum_{a=1}^p \sum_{b=1}^p f\left(e^{\frac{2ia\pi}{p}}, e^{\frac{2ib\pi}{p}}\right)$$

$$= \frac{1}{p^2} \cdot \left(\sum_{a=1}^{p-1} \sum_{b=1}^p f\left(e^{\frac{2ia\pi}{p}}, e^{\frac{2ib\pi}{p}}\right) + \sum_{b=1}^p f\left(1, e^{\frac{2ib\pi}{p}}\right) \right)$$

Since multisets $\{a + b, 2a + b, \dots, 2a + b\} \equiv \{0, 1, 2, \dots, 2p\} \pmod p$ for $\gcd(a, p) = 1$,

$$\sum_{a=1}^{p-1} \sum_{b=1}^p f\left(e^{\frac{2ia\pi}{p}}, e^{\frac{2ib\pi}{p}}\right) = p(p-1) \prod_{j=1}^{2p} \left(1 + e^{\frac{2i(j)}{p}}\right)$$

Claim: $\prod_{j=1}^{2p} \left(1 + e^{\frac{2i(j)}{p}}\right) = 4$ for odd primes p

Note that $\prod_{j=1}^{2p} \left(1 + e^{\frac{2i(j)}{p}}\right) = \left(\prod_{j=1}^p \left(1 + e^{\frac{2i(j)}{p}}\right)\right)^2$. Therefore, we may prove that,

$$\prod_{j=1}^p \left(1 + e^{\frac{2i(j)}{p}}\right) = 2$$

We let $f(x) = x^p - 1$. The desired quantity is $(r_1 + 1) \dots (r_p + 1)$ where $r_1, \dots, r_p$ are the roots of f .

Notice that the product of the roots of $f(x - 1) = (x - 1)^p - 1$ is also $(r_1 + 1) \dots (r_p + 1)$. By Binomial Theorem, the constant term of $f(x - 1)$ is -2 .

By Vieta's since we are given p is odd, $\prod_{j=1}^p \left(1 + e^{\frac{2i(j)}{p}}\right) = 2$ so our quantity is indeed 4.

Using the claim,

$$N = \frac{1}{p^2} \cdot \left(4p(p-1) + \sum_{b=1}^p f\left(1, e^{\frac{2ib\pi}{p}}\right) \right)$$

Now note that,

$$\sum_{b=1}^p f\left(1, e^{\frac{2ib\pi}{p}}\right) = \sum_{b=1}^p \left(1 + e^{\frac{2i(b)}{p}}\right)^{2p}$$

Claim: $\sum_{b=1}^p \left(1 + e^{\frac{2i(b)}{p}}\right)^{2p} = p \binom{2p}{p} + 2p$

We look at the generating function,

$$G(x) = (1 + x)^{2p}$$

The desired quantity is the p roots of unity filter and gives p times the sum of coefficient of the terms with degrees that are multiples of p . By Binomial Theorem, this quantity is $p \left(\binom{2p}{p} + 2 \right)$.

Therefore,

$$\begin{aligned} N &= \frac{1}{p^2} \cdot \left(4p(p-1) + p \left(\binom{2p}{p} + 2 \right) \right) \\ &= \frac{1}{p} \cdot \left(4p - 4 + \binom{2p}{p} + 2 \right) \\ &= \frac{1}{p} \left(\binom{2p}{p} - 2 \right) + 4 \end{aligned}$$

However, this includes the two subsets with 0 or $2p$ elements whose respective sum of elements is indeed a multiple of p , therefore, the actual answer is,

$$N - 2 = \frac{1}{p} \left(\binom{2p}{p} - 2 \right) + 2$$

23 clubs; 8.5 hours

Problem 12 (Shortlist 2021 A2)

We claim the answer is all n where $n + 1$ is prime.

We see,

$$\begin{aligned} & \sum_{i=1}^n \sum_{j=1}^n \left\lfloor \frac{ij}{n+1} \right\rfloor \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{ij}{n+1} - \sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{ij}{n+1} - \frac{1}{2} \left(\sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} + \left\{ \frac{i(n+1-j)}{n+1} \right\} \right) \\ &= \frac{\left(\frac{n(n+1)}{2} \right)^2}{n+1} - \frac{1}{2} \left(\sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} + \left\{ i - \frac{ij}{n+1} \right\} \right) \end{aligned}$$

Therefore,

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^n \left\lfloor \frac{ij}{n+1} \right\rfloor &= \frac{n^2(n-1)}{4} \\
\Leftrightarrow \frac{1}{2} \left(\sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} + \left\{ i - \frac{ij}{n+1} \right\} \right) &= \frac{n^2(n+1)}{4} - \frac{n^2(n-1)}{4} \\
\Leftrightarrow \sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} + \left\{ -\frac{ij}{n+1} \right\} &= n^2
\end{aligned}$$

If $\frac{ij}{n+1}$ is an integer, then $\left\{ \frac{ij}{n+1} \right\} + \left\{ -\frac{ij}{n+1} \right\} = 0$. Otherwise, $\left\{ \frac{ij}{n+1} \right\} + \left\{ -\frac{ij}{n+1} \right\} = 1$.

By Pigeonhole, the zero case may never happen, so $ij \nmid n+1$ for all integers $1 \leq i, j \leq n$. Therefore, $n+1$ must be prime.

Problem 14 (13P4G19)

We see that,

$$\begin{aligned}
\sum_{n \geq 1} \frac{\phi(n) 2^n}{9^n - 2^n} &= \sum_{n \geq 1} \frac{\phi(n) \left(\frac{2}{9}\right)^n}{1 - \frac{2^n}{9^n}} \\
&= \sum_{n \geq 1} \phi(n) \left(\left(\frac{2}{9}\right)^n + \left(\frac{2}{9}\right)^{2n} + \left(\frac{2}{9}\right)^{3n} + \dots \right) \\
&= \sum_{n \geq 1} \sum_{\substack{n|m \\ m > 0}} \left(\frac{2}{9}\right)^m \phi(n) \\
&= \sum_{m \geq 1} \left(\frac{2}{9}\right)^m \sum_{n|m} \phi(n) \\
&= \sum_{m \geq 1} m \left(\frac{2}{9}\right)^m = \sum_{m \geq 1} \left(\frac{2}{9}\right)^m \sum_{k=1}^m 1 = \sum_{k \geq 1} \sum_{m \geq k} \left(\frac{2}{9}\right)^m = \sum_{k \geq 1} \frac{\left(\frac{2}{9}\right)^k}{1 - \frac{2}{9}} = \frac{\frac{2}{9}}{\left(1 - \frac{2}{9}\right)\left(1 - \frac{2}{9}\right)} = \frac{18}{49}
\end{aligned}$$

31 clubs; 10 hours

Problem 16 (18RMM5)

Let a_n be the number of ways to do the problem with $2n$ distinct points labelled $1, \dots, 2n$ in clockwise order.

We also let b_n be the number of ways to do the problem with $2n$ distinct points labelled $1, \dots, 2n$ in clockwise order with the additional condition that all arrows must point from a point with greater label to a smaller label. Note that this means the third condition is never applicable.

We start by solving for b_n .

We casework on where the arrow from point $2n$ points. Notice that if we direct the arrow to a vertex with an odd label, then one side of the arrow would have an odd number of points. Since no arrows intersect, we must pair the remaining points into pairs connected by arrows, which is impossible.

The arrow must point to an even point which we call $2i$ for some $1 \leq i \leq n$. Notice that the points $2, \dots, 2i - 1$ and $2i + 1, \dots, 2n$ may never be connected by an arrow due to the second condition. Therefore, we consider those independently. We may erase points $1, 2i$.

There are b_{i-1} ways to do the points $2, \dots, 2i - 1$ and b_{n-i} ways to do $2i + 1, \dots, 2n$.

$$b_n = \sum_{i=1}^n b_{i-1} b_{n-i} \quad \forall n > 0; b_0 = 1$$

Therefore, by Catalan numbers, $b_n = \frac{1}{n+1} \cdot \binom{2n}{n}$.

We claim that $a_n = 2 \sum_{i=1}^n a_{i-1} b_{n-i} \quad \forall n > 0; a_0 = 1$.

The $n = 0$ case requires no elaboration.

For $n > 0$, we claim there is a bijection between cases where the arrow connecting to point 1 points away from it and to it respectively.

Let f be a function from one valid arrangements of arrows to another that cyclically rotates the numbering until what was originally point 1 is now what was originally on the other side of the arrow connecting to point one. If the domain of f is all cases where the arrow connecting to point 1 points away from it, the range is all cases where the arrow points away from 1. Since $f(f(x)) = x$, f is bijective and we are done.

WLOG, let the arrow direct away from point 1.

We casework on where the arrow from point 1 must be directed to. Notice that if we direct the arrow to a vertex with an odd label, then one side of the arrow would have an odd number of points. Since no arrows intersect, we must pair the remaining points into pairs connected by arrows, which is impossible.

The arrow must point to an even point which we call $2i$ for some $1 \leq i \leq n$.

Notice that no arrows may point from one of points $2, \dots, 2i - 1$ to $2i + 1, \dots, 2n$ or vice versa since no intersections. Therefore, we consider the two sides of our arrow independently.

For points $2, \dots, 2i - 1$, no arrows between those points may couple with the arrow from 1 to $2i$ to trigger condition three. Therefore, we can just solve normally for a_{i-1} ways.

For points $2i + 1, \dots, 2n$, note that if we have any arrow from point $a \rightarrow b$ for $2i + 1 \leq a < b \leq 2n$, then condition 3 will be triggered. Therefore, all arrows must sufficiently and necessarily point from higher labels to lower to prevent condition 3 from being triggered. Therefore, there are b_{n-i} ways.

Therefore, the amount for each case is $a_{i-1}b_{n-i}$ and the total, accounting for the same number of cases where the arrow points to 1 is,

$$a_n = 2 \sum_{i=1}^n a_{i-1}b_{n-i} \quad \forall n > 0; a_0 = 1$$

Since it's Catalan numbers,

$$f(x) = \sum_{n \geq 0} b_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}$$

Note that,

$$\begin{aligned} g(x) &= \sum_{n \geq 0} a_n x^n = 2xf(x)g(x) + 1 \\ \Leftrightarrow g(x) &= \frac{1}{1 - 2x \cdot \frac{1 - \sqrt{1 - 4x}}{2x}} = \frac{1}{\sqrt{1 - 4x}} = \sum_{n \geq 0} \binom{2n}{n} x^n \end{aligned}$$

Therefore, the answer is $\binom{2n}{n}$.

Problem 18 (TWCH1373)

We are given,

$$\frac{1}{1} \left\lfloor \frac{n}{1} \right\rfloor = f_n(1) + f_n(2) + \dots + f_n(n); \quad (\text{equation 1})$$

$$\frac{1}{2} \left\lfloor \frac{n}{2} \right\rfloor = f_n(2) + f_n(4) + \dots + f_n\left(2 \left\lfloor \frac{n}{2} \right\rfloor\right); \quad (\text{equation 2})$$

...

$$\frac{1}{n} \left\lfloor \frac{n}{n} \right\rfloor = f_n(n) \quad (\text{equation n})$$

We start with the first equation and take $n - 1$ steps as follows. We start with,

$$\frac{1}{1} \left\lfloor \frac{n}{1} \right\rfloor = f_n(1) + f_n(2) + \dots + f_n(n)$$

For the first step, we see that the coefficient of $f_n(2)$ is 1, so we add -1 times the second equation to both sides to get,

$$\frac{1}{1} \left\lfloor \frac{n}{1} \right\rfloor - \frac{1}{2} \left\lfloor \frac{n}{2} \right\rfloor = f_n(1) + f_n(3) + f_n(5) + \dots$$

We continue the process where at the i -th step, we look at the current coefficient of $f_n(i + 1)$, which we call x_{i+1} . We then add $-x$ of the $(i + 1)$ -th equation to both sides.

For completeness, we let $x_1 := 1$. After $n - 1$ such steps, we get,

$$x_1 \frac{1}{1} \left\lfloor \frac{n}{1} \right\rfloor + x_2 \frac{1}{2} \left\lfloor \frac{n}{2} \right\rfloor + x_3 \frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor + \cdots + x_n \frac{1}{n} \left\lfloor \frac{n}{n} \right\rfloor = f_n(1)$$

We claim that $x_i = \mu(i)$ the mobius function. We prove this by induction,

Base case: $i = 1, x_1 = \mu(1) = 1$

Inductive step: $i = k; i > 1$

We look at the coefficient of $f_n(k)$ before the start of the $(k - 1)$ -th step. We see that, $f_n(k)$ appears once for every time we add equation j over all j that are divisors of k . For the case where $j = 1$, we can imagine there is a 0-th step where we added the first equation to both sides of $0 = 0$.

$$x_k = -\text{current coefficient of } f_n(k) = -\sum_{\substack{j|k \\ j \neq k}} x_j = -\sum_{\substack{j|k \\ j \neq k}} \mu(j)$$

Since we know $\sum_{j|k} \mu(j) = 0$ for $i > 1$, add zero to both sides to get,

$$\Leftrightarrow x_k = \mu_k$$

Conclusion: By induction we are done.

Therefore,

$$f_{1000}(1) = \sum_{k \leq 1000} \mu(k) \cdot \frac{1}{k} \cdot \left\lfloor \frac{1000}{k} \right\rfloor;$$

$$f_{999}(1) = \sum_{k \leq 999} \mu(k) \cdot \frac{1}{k} \cdot \left\lfloor \frac{999}{k} \right\rfloor;$$

Note that, $\left\lfloor \frac{1000}{k} \right\rfloor - \left\lfloor \frac{999}{k} \right\rfloor$ is 1 if $k \mid 1000$ and 0 otherwise. When $k \mid 1000$, $\left\lfloor \frac{1000}{k} \right\rfloor$ is an integer and we know $\frac{1000}{k} - \frac{999}{k} \leq 1$.

In the other case, we can check that there is never an integer less than $\frac{1000}{k}$ and greater than $\frac{999}{k}$. If there is,

$$\frac{1000}{k} > n; n > \frac{999}{k}$$

$$\Leftrightarrow 1000 > kn > 999$$

This is impossible for integers k, n .

Therefore,

$$f_{1000}(1) - f_{999}(1) = \sum_{k|1000} \mu(k) \cdot \frac{1}{k} = 1 - \frac{1}{2} - \frac{1}{5} + \frac{1}{10} = \frac{2}{5}$$

37 clubs; 12.5 hours